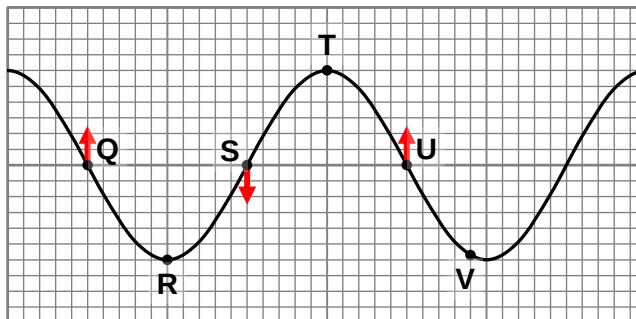


5 (a)



(b) Particles R and T are momentarily at rest.

I (i) in phase: any pair of points a wavelength apart

(ii) In antiphase: any pair of points half-wavelength apart

(d) Particle U leads particle V by a phase of

$$\frac{\Delta x}{\lambda} \times 2\pi = \frac{4 \text{ small squares}}{20 \text{ small squares}} \times 2\pi = 0.4\pi \text{ rad}$$

Therefore, V leads U by a phase of $\phi = 2\pi - 0.4\pi = 5.0265 = 5.03 \text{ rad}$

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